WORKSHEET · SCRATCH (GR 3–5) INTERMEDIATE: STORY, QUIZ & MUSIC STUDIO

Lesson 1: Loops Inside Loops: Pattern Art

Module 1 | 45 minutes | Nested loops | Grades 3–5 · ages 8–10

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

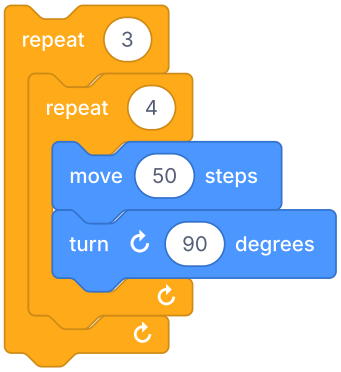
a. nested loop ___ A. How thick the pen line is.
b. inner loop ___ B. The loop on the outside that repeats the inner loop.
c. outer loop ___ C. The loop inside; it runs completely each time.
d. pen size ___ D. A loop inside another loop.

2. Plan your code

Before you build it: Draw a flower: 8 squares, turning 45 degrees after each square. Use a loop inside a loop. Write or draw the blocks you will use.

3. Count the work

How many times does the move block run? Circle one.



a. A. 12 times
b. B. 7 times
c. C. 4 times

4. Loop maths

An outer loop of 6 with an inner loop of 5 runs the inner blocks _____ times. To spread 6 shapes around a circle, turn _____ degrees after each one.

5. Plan your code

Before you build it: Draw a row of 3 squares side by side. Use a nested loop, and lift the pen to move to the next square. Write or draw the blocks you will use.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can build a loop inside a loop.
☐ I can work out how many times the inner blocks run.
☐ I can space shapes evenly around a circle.